

Extending Agile Processes with Creativity Techniques

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// Agile processes seek “just enough” requirements, but a real-world case study shows that this approach can come at the expense of more creative requirements. //

ALTHOUGH ADVOCATES OF agile claim that it can deliver innovative requirements and software in uncertain situations,¹ concrete evidence for this claim is lacking. Indeed, agile processes make little reference to established creativity theories and techniques, and some researchers now question whether agile processes can deliver innovation.²

One obvious weakness is that agile processes don't exploit working software for creative thinking about new requirements. Indeed, the short durations of sprints can discourage the incubation and reflection needed for creative thinking.³ Often the principle of

simplicity—eliminating waste and reducing complexity—is taken too far. For example, Kent Beck advises agile developers to think of the simplest solution.⁴ Although this approach is often effective, unless a concerted effort is made to see the potential beyond the simplest solution, customers will settle for solutions developed in ignorance of what's possible. This limitation is reinforced by agile's adoption of data-driven techniques that focus on current practices such as competitor analysis.⁵

Integrating creative thinking about requirements more effectively into agile processes is a challenge. Here, we de-

scribe the extension of one real-world agile project with creativity techniques and an evaluation of the results. The agile project belonged to BBC Worldwide, the wholly owned commercial subsidiary of the UK's principal public service broadcaster.

Framing Agile Requirements for Creativity

We reviewed agile processes to determine where and how to inject creativity techniques into them. We adopted Robert Sternberg's definition of creativity as “the ability to produce work that is both novel (i.e. original, unexpected) and appropriate (i.e. useful)...”⁶ So we wanted to inject techniques that would generate novel and useful requirements. By useful, we sought requirements that, if satisfied by the software solution, would contribute positively to achieving one or more project-specified goals.

We based our solution on part of Scott Ambler's agile process,⁷ extending it in two places with the creative activities (see Figure 1):

- An extended envisioning process during sprint zero to discover a novel system vision and high-level requirements, known as *epics*.
- An extended epic process to both select epics with more potential for creative outcomes and support the discovery of novel requirements for each one.

To explore which creativity techniques to implement in each new creativity stage, we framed the envisioning and epic processes using the stages and techniques of version 6.1 of the Creative Problem Solving (CPS) process defined by Scott Isaksen and his colleagues.⁸ The CPS process provides

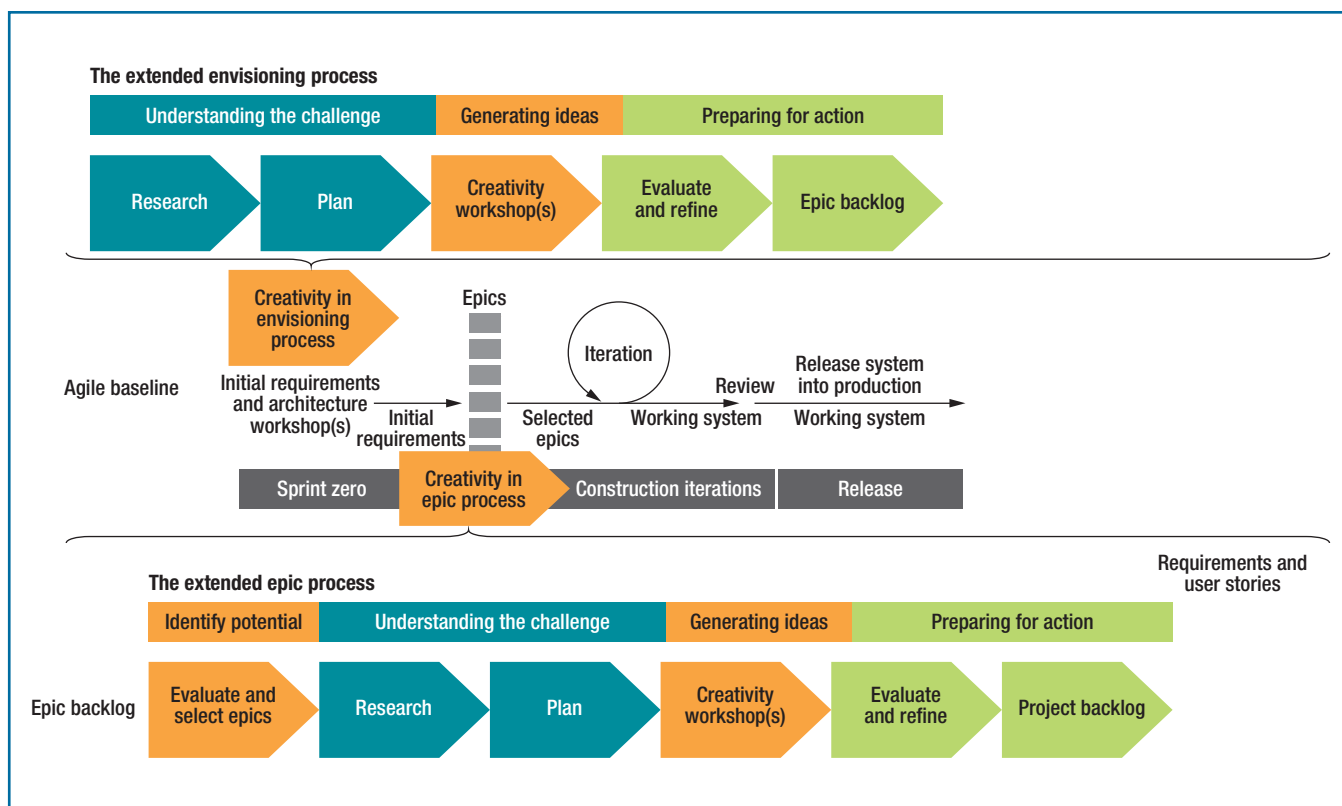


FIGURE 1. The extended agile process. The envisioning process and an iteration of the epic process extend the agile baseline to introduce two creativity activities. The extended processes follow the Creative Problem Solving (CPS) process.⁸ The creativity activities and extended process stages appear in orange.

comprehensive guidance for individuals, teams, and organizations to treat people, content, and context as part of a problem-solving system. The CPS process stages are divided into three phases:

- understanding the challenge,
- generating ideas, and
- preparing for action,

with an optional preliminary phase, identifying potential, to discover challenges that can benefit from creativity techniques.

Key stages within the three main phases include constructing opportunities, exploring data, framing the prob-

lem, generating ideas, developing solutions, and building acceptance.

In our framing, some CPS and agile process stages are equivalent. Exploring data can map to the use of agile data analysis techniques, such as *personas* that act as proxy stakeholders for exploring a domain when stakeholders aren't directly available. Likewise, framing the problem can map to the development of epics and user stories in sprint zero and beyond. Developing the solution can map to model storming in early sprints, and building acceptance can map to later sprint activities such as the iterative reviews, the working software with customers, and the system's release into production.

However, mapping agile techniques to the CPS stage for generating ideas was more difficult. Initial requirements and architecture workshops offer little support for creating requirements—a weakness exacerbated by the short durations of sprints. Therefore, generating ideas became the focus of our new process. We extended the envisioning and epic processes with the explicit use of idea generation techniques.

The extended envisioning process (see Figure 1) defines the vision for a product and the production of epics using the three CPS phases. Generating ideas is a new phase for the agile process. It implements one or more short creativity workshops to discover novel

ideas to inform the vision and epics. For the BBC case study, we used established creativity techniques, such as those defined by Michael Michalko⁹ and James Higgins.¹⁰

The extended epic process (see Figure 1) discovers new ideas that become

one meeting takes place to evaluate the requirements and identify the ones to move forward. We recommend one week between the workshop and the evaluation to provide a crucial incubation period for the requirements.

The final step is to present details

We investigated H2 because previous studies of creativity support for requirements work revealed a trade-off between requirement novelty and usefulness.¹²

We also wanted to investigate whether the adding the idea-generation phase resulted in outputs that were more novel than the epic process inputs. One alternative that we considered but rejected was to compare the requirements generated using the original and extended epic processes, but this would have introduced new uncontrolled variables arising from our inability to isolate the original epic process from the effects of the extended one.

The extended epic process discovers new ideas that become the baseline requirements in the product backlog.

the baseline requirements in the product backlog. This process also implements the three CPS phases, which we designed around generating ideas through one or more short creativity workshops to discover novel requirements from one epic at a time. Furthermore, the themes of these creativity workshops are determined in the new preliminary phase to identify potential novel requirements. During this phase, the team evaluates existing epics in the product backlog to select one or more with potential for leading to creative requirements. The team uses four CPS criteria for epic selection,⁸ which we adapted a little to the agile process:

- The epic is ambiguous.
- The epic is complex.
- The epic necessitates novelty to offer competitive advantage.
- The epic offers openness for creative potential.

For selected epics, the extended process then runs one or more creativity workshops during sprints to create new outline requirements and user stories on the future system, informed by research and planning stages similar to the envisioning process. During the evaluate-and-refine stage, at least

of the extended envisioning and epic processes to project management for evaluation.¹¹

Evaluating the Extended Epic Process

At BBC Worldwide, we collected evidence to investigate whether the extended epic process, under real-world time constraints, could lead to requirements that were more novel than the baseline epics in the agile project's backlog.

We evaluated several projects, but we report only one here—namely, the redesign of a television listings website. We evaluated the project by comparing ratings made by independent domain experts about the novelty and usefulness of the requirements from both sources. We applied statistical analyses to the ratings to explore two hypotheses:

- H1: Requirements generated from applying the extended epic process were rated more novel than epics in the original product backlog for the same project.
- H2: Requirements generated from applying the extended epic process were rated less useful than epics in the original product backlog for the same project.

The Television Listings Website

The product owner and key stakeholders had already produced a clear product vision and backlog of epics for the television listings website. The competitor analysis had identified the need for capabilities that let users personalize their website access and become a member of a community—equivalent to being able to connect with friends via social networking sites. The BBC agile project had balanced these epics with the product's core values and unique selling points to produce a product proposition statement and backlog of 216 epics. At that point, a team composed of the lead developer, the head of design, a business analyst, and a second developer agreed to use our extended epic process with facilitation from the first author. The team had formed six weeks earlier, and each member had between three and six years' experience in agile projects.

The original agile process had combined standard practices derived from the Agile Manifesto. For example, it had applied Kanban to visualize work progress, Scrum to provide regular feedback cycles, XP development practices to control software quality, and

automated tools to document software. We considered the process to be typical of agile projects.

The Epic Process Rollout

The case study followed the six stages of the extended epic process (see Figure 1) over two weeks. During the identify potential stage, the project team selected which epic had the greatest potential for creative exploitation, according to the four CPS characteristics of ambiguity, complexity, novelty for competitive advantage, and openness for creative potential. The team then used the selected epic to generate a challenge statement for a creativity workshop:

In what ways might the system change a user's profile based on their behavior inside or outside of the website?

The research and planning stages then took place over five days to prepare for one creativity workshop that ran for 60 minutes with the same head of design, business analyst, and two developers, all of whom had knowledge of the domain and epics backlog. The workshop participants acted as proxy users because the agile project's time-scale wouldn't permit the participation of real end users in the research.

At the workshop's start, the facilitator prepared the four participants for creative thinking. She communicated and clarified the challenge statement to ensure that the participants shared a common starting point. The rules for creativity were set:

- respect, acknowledge, and build on the ideas of others;
- avoid negative feedback and comments;
- generate as many as ideas as possible; and
- don't think about how to implement the ideas.

A short, fun task was undertaken together to break the ice among team members.

The workshop then used three creativity techniques: brainstorming; random stars, a new technique inspired by Michalko's hall-of-fame technique;⁹ and Higgins's Partners in Creative Learning (PICL) technique.¹⁰ The facilitator documented ideas on large Post-it Notes to support participants' sketching and drawing.

The brainstorming technique was familiar to the workshop participants, but it hadn't been used effectively within the department. The facilitator orchestrated an open brainstorm that allowed participants to share their favorite ideas and other participants to build on them to generate requirements.

For the random-stars technique, we adapted the original hall-of-fame technique to television programs to make it fun and relevant. The facilitator chose

domly selected one statement at a time and used it to discover new website requirements. Again, participants used the statements to force the generation of new website requirements based on the statement themes.

During the evaluate-and-refine stage, a one-week period of incubation allowed more creative thinking about the requirements generated during the workshop. The project team met once to review workshop outcomes and decide which requirements to move forward.

Workshop Results

The creativity workshop lasted 55 minutes, during which the four participants generated 40 new requirements—30 with the brainstorming technique and 10 with the random-stars and PICL techniques.

The brainstorming session led to requirements that involved novel uses of other websites and technologies. One

The case study followed the six stages of the extended epic process over two weeks.

14 BBC programs, represented by their main characters—for example, Vicky Pollard from *Little Britain*, Alan Sugar from *The Apprentice*, and the *Top Gear* presenters. Photographs of each character were printed out, cut up, and placed in a bag. Participants used the characters to force connections and generate new website requirements based on themes.

The photographs were supplemented with 25 short statements based on the PICL technique and designed to trigger new lines of thought—for example, “a television program all about you,” “make it like ants,” and “add alcohol.” For this technique, participants ran-

requirement described a chat-roulette feature that would let users digitally drop in on friends to see what they're watching. Another described a feature to view people's reactions to a television program through a webcam. Participants produced these requirements at a rate of one per minute, and the facilitator observed that a moment rarely passed without someone talking about a requirement.

The random-stars and PICL session led to ideas associated more with emotional connections to programs. One requirement described a capability to capture a user's emotional response to a program or character. Another

described a capability that let users indicate which actors they lusted for. A third used the characters as personas—for example, Vicky Pollard led to an idea that missing a television program would leave a viewer a social leper. The participants generated the 10 random-

We limited the comparison to 40 requirements and epics. We thought it reasonable to expect a domain expert to accurately rate this total number in one session lasting 45 minutes. The backlog contained 216 epics and the extended epic process generated 40

Ten volunteer domain experts independently rated the novelty and usefulness of 40 requirements and epics.

stars and PICL requirements more slowly than requirements, due in part to the need to draw prompt items out of the bag, read them, and find connections with the website.

Two days after the workshop, during the evaluate and refine stage, the lead developer and business analyst from the workshop presented the 40 requirements to the product owner using the original Post-it Notes. At the time, the product owner showed greatest interest in two requirements—a colored mood board and the site as a personality—but neither were selected to implement. However, one week later, after more incubating of the requirements, the agile project team reported that the workshop had influenced the product owners' thinking and, as a result, changed their requirements on the website.

Post-Workshop Analysis

We used domain expert ratings of the requirements from the extended epic process and epics in the original epic backlog to investigate our hypotheses, H1 and H2. We used the same number of individual requirements from the creative epic process and epics from the original project's epic backlog to generate two datasets that the domain experts then rated independently for their degrees of novelty and usefulness.

We discounted 85 epics that requested features common to most websites, then randomly picked 20 epics from the rest, along with 20 requirements from the 40 generated in the extended epic process.

We transcribed the 20 epic-process requirements from the Post-it Notes and workshop audio recordings and described each one in 20 to 50 words. This required the omission and addition of some words to capture the essence of the idea in the short format, noted by the ellipsis (...) and square brackets ([])—for example, “we [could] see when people have searched for film times... and what kind of films they are interested in and then... feed[ing] that data back into their preferences, ... we could tell them when it's [showing] on TV.”

The original agile process team had documented the backlog epics as more concise, factual statements than the requirements generated from the extended epic process. To remove this potential source of bias, we presented both sets in the same format. We asked one independent analyst to describe the 20 selected backlog epics in a similar fashion to the requirements generated in the extended epic process. We didn't tell the analyst how the data would be used but asked her to describe each requirement with passion and to talk

in the future tense, as if she had just thought of the requirement. We then used the same process we used for the epic-process requirements to transcribe and condense the epics into 20- to 50-word descriptions, thereby ensuring equivalent style and length to the requirements from the extended epic process.

We listed the selected requirements and epics in random order in a questionnaire with a numerical scale of 1 to 5 for rating novelty and usefulness of each one. The questionnaire used the following definitions:

- *Novelty*. In the context of a television listings website, the idea is original, imaginative, or unique.
- *Usefulness*. In the context of a television listings website, the idea is of some potential value to the website or its users. For example, a game might not be “useful” in the classic sense but it might add some commercial or user value in the right context.

Ten volunteer domain experts not associated with the television listings website project then independently rated the novelty and usefulness of the 40 requirements and epics in the questionnaire.

H1 Results

We applied an unpaired, two-tail, unequal variance t-test to the 400 novelty ratings collected from the 10 domain experts. The average novelty rating given to the epics from the backlog was 2.69 out of 5, and the average novelty rating given to requirements from the extended epic process was 3.28 out of 5. A statistical analysis revealed that requirements generated from the extended epic process were rated as significantly more novel than epics in the original product backlog. The result supported hypothesis H1.

Moreover, five of the 20 requirements from the extended epic process received an average rating of 3.9 or higher out of 5. The following requirement received the highest rating of 4.7: “The [site] should serve up content based on what you’re watching now, for example, ‘Here’s Seasons 1–5 available now on DVD or video on demand... so [if you want to, you can] start right from the start.’”

The high rating suggested a potential killer requirement that transforms a new application into an innovative one. The extended epic process led the team to generate some killer requirements, some requirements that were novel and could lead to innovative change, and some requirements that weren’t novel.

H2 Results

We also applied an unpaired, two-tail, unequal variance t-test to the 400 usefulness ratings. The epics in the original product backlog received an average usefulness rating of 3.37 out of 5, and the extended epic process’s requirements received an average novelty rating of 2.97 out of 5. A statistical analysis revealed that epic-process requirements were rated as significantly less useful than epics in the original product backlog. The result supported hypothesis H2.

Threats to Validity

Of course, these results are subject to potential validity threats.¹³ With respect to conclusions, the extended epic process was open to other project influences, although this threat was mitigated by the process’s short duration, during which personnel and other key factors didn’t change.

Threats to internal validity include the possibility that using the extended epic process might have increased stakeholder motivation to be more creative, independent of the process stages.

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However, the domain experts reviewed the new requirements as documented at the creativity workshop’s conclusion, so this effect on motivation would be limited to the stages of identifying potential, understanding the challenge, and generating ideas.

Another obvious threat to internal validity was our decision not to compare the new requirements to requirements generated directly during the original epic process. However, the project team had previously applied the original process to some epics, and outcome requirements from that effort had fed back into the product backlog, so to some extent, our comparison against epics in the product backlog did take these requirements into consideration.

A third threat was the use of just one independent analyst to author all 20 selected backlog epics. We acknowledge this threat and sought to mitigate it with clear guidelines about how to write each epic in a form similar to the new requirements.

One threat to external validity was that the results reported here come from just one agile project. We also ac-

knowledge this. However, given the absence of published empirical data about creativity in agile projects, our aim was to demonstrate feasibility rather than make general claims. Qualitative data from extending a second agile project at BBC Worldwide did support the results from this project,¹¹ and other projects are currently using the extended agile processes. We also acknowledge that the sample of 40 requirements and epics might not represent all the requirements. The size was constrained by the number of ideas that one domain expert could rate reliably in an experimental setting.

Lan Cao and Balasubramaniam Ramesh reported evidence that agile processes enabled customers to steer projects to meet unanticipated requirements.¹⁴ In our view, this capability is necessary but not always sufficient to generate the requirements for innovative software solutions. Our preliminary research shows that selecting an epic with creative potential and deliberately generating

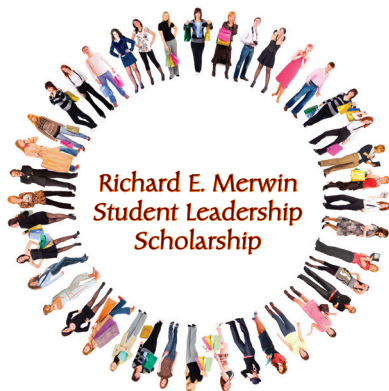
ideas during a sprint can produce more requirements ranked as more novel. Although these requirements initially seemed less useful, giving a week to allow subsequent idea incubation during the sprint appeared to increase their perceived potential usefulness. Our initial research suggests that agile projects can reach beyond the simplest solution without the costs often associated with upfront creativity workshops.¹⁵ It provides preliminary evidence that people can deliver creativity on a shoestring in agile projects. That said, not all agile projects will be seeking more novelty at the expenses of usefulness. Therefore, we recommend

that creativity techniques be used in projects seeking step—rather than incremental—change.

So what next? We want more agile practitioners to use lightweight creativity techniques. So if you want to participate in this work, let us know. ☺

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